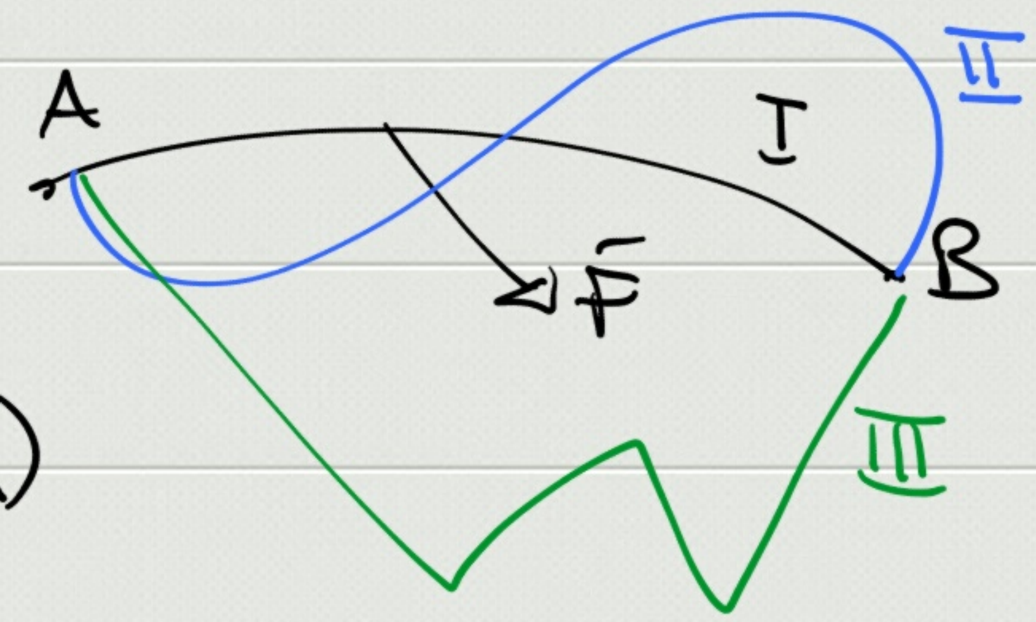
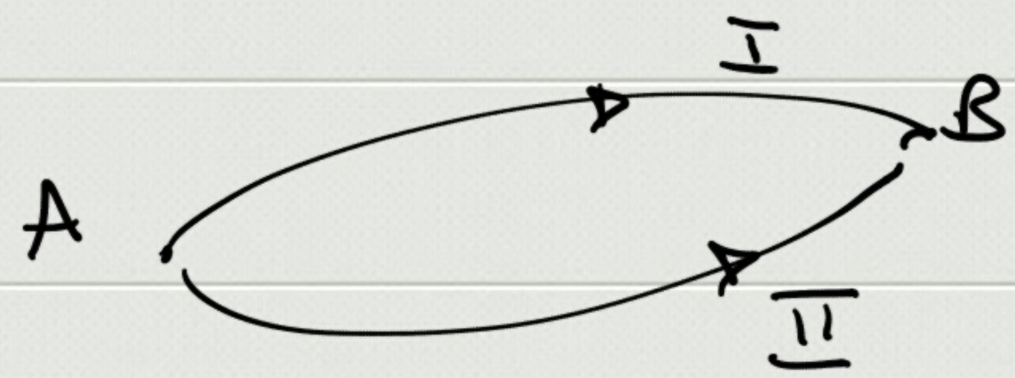
$$W_{A\to B} = \int_A^B \bar{F}\, d\bar{s} = -\Delta E_p =$$

$$= -\left(E_{p,B} - E_{p,A}\right)$$



$$W_{A\to B} = \int_A^B \left(\bar{F}\, d\bar{s}\right)_{I} = \int_A^B \left(\bar{F}\, d\bar{s}\right)_{II} = \int_A^B \left(\bar{F}\, d\bar{s}\right)_{III} = -\Delta E_p$$

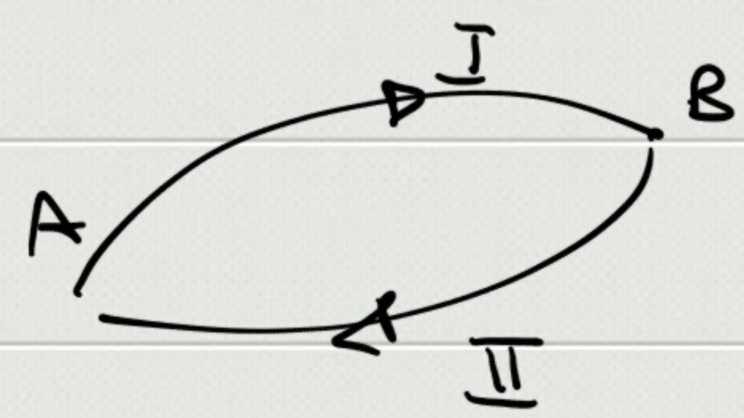


$$\int_A^B \left(\bar{F}\, d\bar{s}\right)_{I} = \int_A^B \left(\bar{F}\, d\bar{s}\right)_{II}$$

$$\int_A^B \left(\bar{F}\, d\bar{s}\right)_{I} - \int_A^B \left(\bar{F}\, d\bar{s}\right)_{II} = 0 \qquad *$$

$$W_{A\to B} = \int_A^B \bar{F}\, d\bar{s} = -\left(E_{p,B} - E_{p,A}\right) = -\left[-\left(E_{p,A} - E_{p,B}\right)\right] =$$

$$= -\int_B^A \bar{F}\, d\bar{s} = -W_{B\to A}$$



$$\int_A^B \left(\bar{F}\, d\bar{s}\right)_{I} + \int_B^A \left(\bar{F}\, d\bar{s}\right)_{II} = 0$$

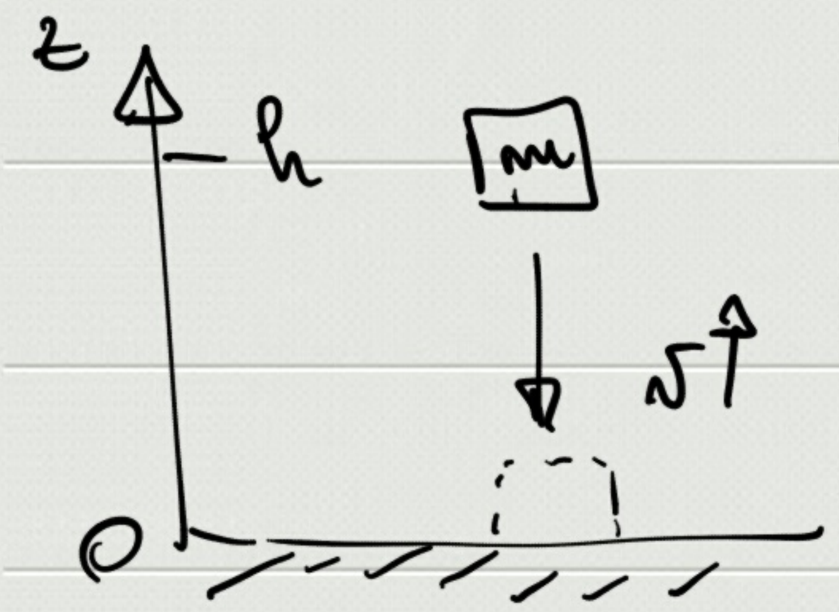
$$\boxed{\oint \bar{F}\, d\bar{s} = 0}$$

(circuitazione)

forze conservative

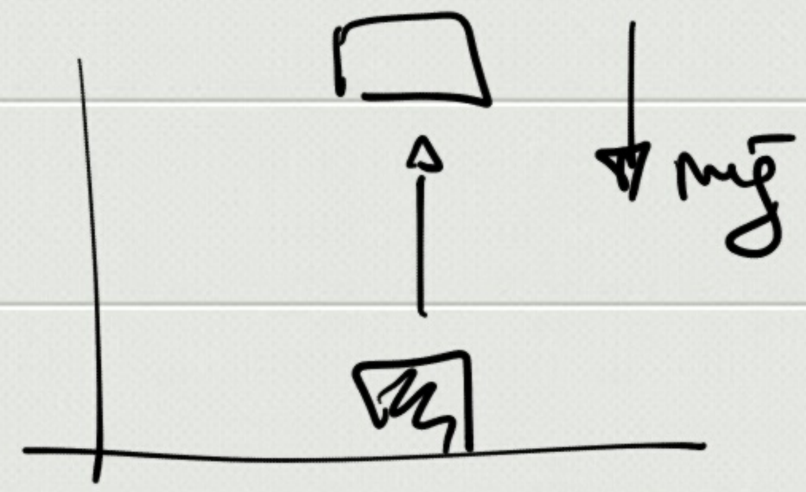
Forze conservative $\boxed{W_{A\to B} = -\Delta E_p} \Rightarrow \oint \bar{F}\, d\bar{s} = 0$

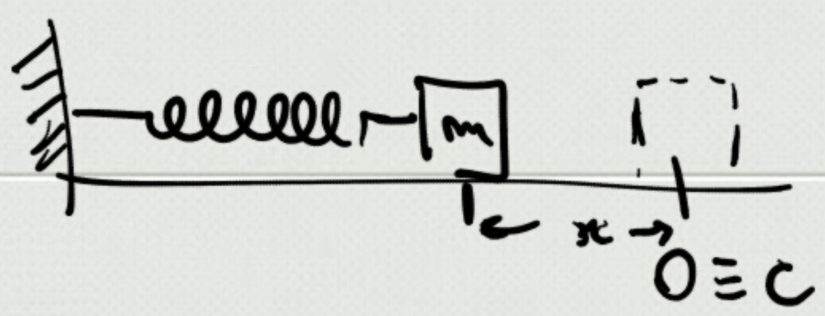
$$W_{A\to B} = \underset{\text{iniziale}}{E_{p,A}} - \underset{\text{finale}}{E_{p,B}} \begin{cases} >0 : \text{motore} \Rightarrow E_{p,in} > E_{p,fin} \\ <0 : \text{resistente} \Rightarrow \text{"} < \text{"} \end{cases}$$



$E_p = mgh$

$E_p = 0$



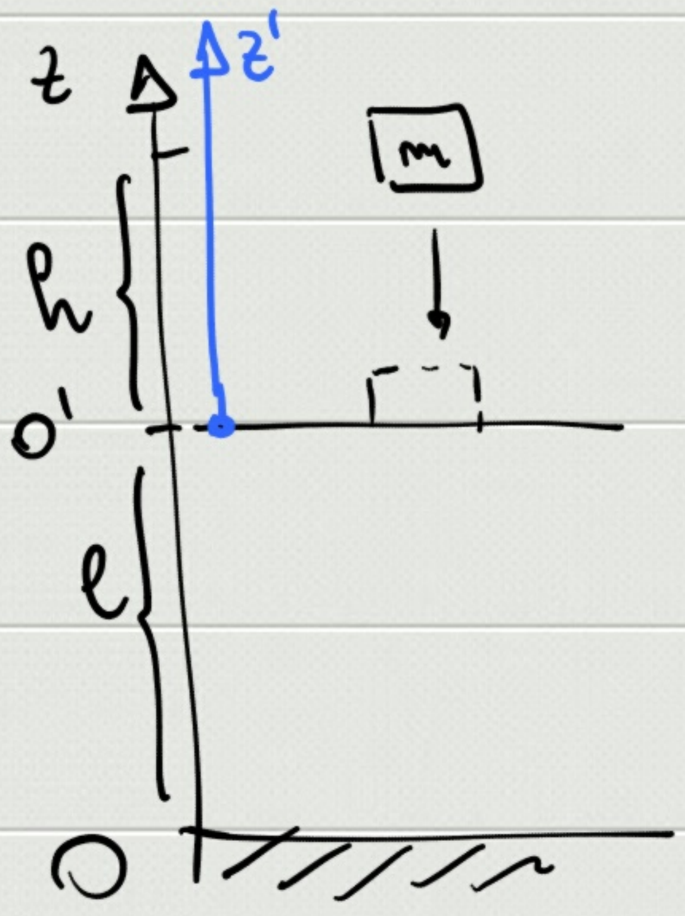


$E_p = \frac{1}{2} k x^2$

$$W_{A\to B} = -\Delta E_p = -(E_{p,B} - E_{p,A})$$

$$E_{p,peso} = mgz + cost$$

$$\Uparrow$$



$E_p = mg(l+h)$ $\quad E'_p = mgh$

$E_p = mgl$ $\quad E'_p = 0$

$\Delta E_p = -mgh$

$E'_p = mgz - mgl$

Forze conservative : $W = -\Delta E_p$

Forze non conservative : $\cancel{W = -\Delta E_p}$

(dissipative)

$$\left.\right\} \boxed{W_{A\to B} = \Delta E_k}$$

Forze conservative

$$\left.\begin{aligned} W_{A\to B} &= -\Delta E_p = -(E_{p,B} - E_{p,A}) \\ &\overset{!}{=} \Delta E_k = E_{k,B} - E_{k,A} \end{aligned}\right\} \Rightarrow$$

$$E_{k,B} + E_{p,B} = E_{k,A} + E_{p,A}$$

$$\boxed{E_m = E_k + E_p} \rightarrow \text{energia meccanica}$$

$$\Rightarrow \boxed{E_{m,B} = E_{m,A}}$$

Principio di conservazione dell' energia meccanica

Forze non conservative

$$\Rightarrow \quad W_{TOT,\,A\to B} = W_{m.c,\,A\to B} + W_{fcons.\,A\to B} =$$

$$\left.\begin{aligned} &= W_{mc} - \Delta E_p \\ &\overset{!}{=} \Delta E_{k,\,A\to B} \end{aligned}\right\} \Longrightarrow$$

$$W_{mc} = \Delta E_k + \Delta E_p = \Delta(E_k + E_p) = \Delta E_m$$

$$\boxed{W_{mc,\,A\to B} = E_{m,B} - E_{m,A}}$$